

1 The Jordan–Wigner mapping for spinless Fermions

1.1 Summary

The Jordan–Wigner transformation (JWT) can be used to map a fermionic system to a system of spins. This is essentially done by rewriting

$$\text{fermion} = \text{string} \times \text{operator},$$

where the “string” is some product of operators $\hat{Z}_i = 1 - 2\hat{n}_i$ and the “operator” is found to obey spin commutation relations.

As references I used

- *Fermions and Jordan–Wigner String* $\sim$ ITensor C++ Documentation,
- *Introduction to Many-Body Physics*, Chapter 4.1 $\sim$ Piers Coleman,
- *The Fermionic canonical commutation relations and the Jordan–Wigner transform* $\sim$ M. Nielsen,

which I supplemented with some calculations and my own way of structuring things.

1.2 Setup: the fermionic Hilbert space

Our fermionic Hilbert space is defined by (Nielsen):

- i) operators $c_1, \dots, c_L$ acting on some Hilbert space;
- ii) these operators satisfying the canonical (anti-)commutation relations for fermions

$$\{c_i^{(+)}, c_j^{(+)}\} = 0 \quad \& \quad \{c_i, c_j^\dagger\} = \delta_{ij} \mathbb{1}. \quad (1)$$

Many things directly follow from this, such as a possible basis for the Hilbert space, namely the occupation number basis

$$|\vec{n}\rangle = |n_1, n_2, \dots, n_L\rangle \equiv \left(\prod_{i=1}^L (c_i^\dagger)^{n_i} \right) |0\rangle, \quad (2)$$

where $\hat{n}_i |\vec{n}\rangle = c_i^\dagger c_i |\vec{n}\rangle = n_i |\vec{n}\rangle$ with $n_i = 0, 1$ and

$$[\hat{n}_i, \hat{n}_j] = 0. \quad (\star_1)$$

1.3 The transformation

The JWT introduces new operators a_j by writing

$$c_j \equiv \left(\prod_{i=1}^{j-1} Z_i \right) a_j \quad \& \quad c_j^\dagger \equiv a_j^\dagger \left(\prod_{i=1}^{j-1} Z_i \right), \quad (*)$$

with $Z_j = (-1)^{n_j} = 1 - 2n_j$ and $n_j = c_j^\dagger c_j$. The ordering of the Z_i does not matter since $[Z_i, Z_j] = 0$ (using $(\star_1)$).

Inverting $(*)$ using $Z_i^2 = \mathbb{1}$ and $[Z_i, Z_j] = 0$ yields

$$a_j = \left(\prod_{i=1}^{j-1} Z_i \right) c_j \quad \& \quad a_j^\dagger = c_j^\dagger \left(\prod_{i=1}^{j-1} Z_i \right). \quad (*')$$

1.4 Algebra of the new operators

One can show that these new operators obey

$$[a_i^{(+)}, a_j^{(+)}] = 0 \quad \forall i, j, \quad (3)$$

$$[a_i, a_j^\dagger] = 0 \quad \forall i \neq j, \quad (4)$$

$$[a_i, a_i^\dagger] = 1 - 2n_i \quad \text{with } n_i = c_i^\dagger c_i, \quad (5)$$

using that

$$\{Z_i, c_j^{(+)}\} = 0 \quad (\star_2)$$

(proofs further below).

1.5 The occupation-number basis in the new operators

The basis states of the fermionic Hilbert space can be written as

$$\begin{aligned} |n_1, n_2, \dots, n_L\rangle &\equiv \left(\prod_{i=1}^L (c_i^\dagger)^{n_i} \right) |0\rangle && \text{w. } n_i = 0, 1 \\ &= \left\{ \prod_{i=1}^L \left[a_i^\dagger \left(\prod_{j=1}^{i-1} Z_j \right) \right]^{n_i} \right\} |0\rangle && (c_1^\dagger = a_1^\dagger) \\ &= (a_1^\dagger)^{n_1} (a_2^\dagger Z_1)^{n_2} (a_3^\dagger Z_2 Z_1)^{n_3} \dots (a_L^\dagger Z_{L-1} \dots Z_1)^{n_L} |0\rangle \\ &= \left(\prod_{i=1}^L (a_i^\dagger)^{n_i} \right) (Z_1)^{n_2} (Z_2 Z_1)^{n_3} \dots (Z_{L-1} \dots Z_1)^{n_L} |0\rangle \\ &= \left(\prod_{i=1}^L (a_i^\dagger)^{n_i} \right) |0\rangle && \text{w. } n_i = 0, 1, \end{aligned} \quad (6)$$

using $[Z_i, Z_j] = 0 \ \forall i, j$, $[Z_i, c_j] = 0$ for $i \neq j$ (hence $[Z_i, a_j^\dagger] = 0$ for $i \neq j$), and $Z_j |0\rangle = |0\rangle$. The ordering is not important anymore because $[a_i^\dagger, a_j^\dagger] = 0$.

1.6 Action on basis states

One cannot define the action of a_j ($a_j^\dagger$) locally, since the JW string $Z_1 \cdots Z_{j-1}$ is non-local, so one has to examine them on the full many-body basis. For $n_\ell = 0$,

$$\begin{aligned} a_\ell |n_1, \dots, n_\ell = 0, \dots, n_L\rangle &= a_\ell \left(\prod_{i=1}^L (a_i^\dagger)^{n_i} \right) |0\rangle \\ &\stackrel{[a_\ell, a_j^\dagger]=0}{=} \left(\prod_{i=1}^L (a_i^\dagger)^{n_i} \right) a_\ell |0\rangle = 0, \end{aligned} \quad (7)$$

since $a_\ell |0\rangle = c_\ell |0\rangle = 0$. For $n_\ell = 1$,

$$\begin{aligned} a_\ell |n_1, \dots, n_\ell = 1, \dots, n_L\rangle &= a_\ell \left(\prod_{\substack{i=1 \\ i \neq \ell}}^L (a_i^\dagger)^{n_i} \right) a_\ell^\dagger |0\rangle \\ &\stackrel{[a_\ell, a_j^\dagger]=0}{=} \left(\prod_{\substack{i=1 \\ i \neq \ell}}^L (a_i^\dagger)^{n_i} \right) a_\ell a_\ell^\dagger |0\rangle = \left(\prod_{\substack{i=1 \\ i \neq \ell}}^L (a_i^\dagger)^{n_i} \right) c_\ell c_\ell^\dagger |0\rangle \\ &= \left(\prod_{\substack{i=1 \\ i \neq \ell}}^L (a_i^\dagger)^{n_i} \right) |0\rangle = |n_1, \dots, n_\ell = 0, \dots, n_L\rangle. \end{aligned} \quad (8)$$

Using $[a_\ell^\dagger, a_j^\dagger] = 0$ and $(a_\ell^\dagger)^2 = 0$, one quickly finds

$$a_\ell^\dagger |n_1, \dots, n_\ell = 0, \dots, n_L\rangle = |n_1, \dots, n_\ell = 1, \dots, n_L\rangle, \quad (9)$$

$$a_\ell^\dagger |n_1, \dots, n_\ell = 1, \dots, n_L\rangle = 0. \quad (10)$$

Thus a_ℓ ($a_\ell^\dagger$) destroys (creates) a particle in the many-body state *without* giving rise to an extra sign structure (as c_ℓ ($c_\ell^\dagger$) would do).

1.7 Identification with spins

We can thus identify $n_i = 0, 1$ with $n_i = \downarrow, \uparrow$ in the many-body state, and the new operators can be associated with spins by identifying

$$a_j = S_j^-, \quad a_j^\dagger = S_j^+ \quad \& \quad n_j - \frac{1}{2} = S_j^z = \frac{1}{2} \sigma_j^z, \quad (11)$$

since $[S_j^-, S_j^+] = -2S_j^z$ for spins (see the quick refresher on spin, §??). For instance

$$a_3^\dagger |0\rangle = S_3^+ |\downarrow\downarrow\downarrow\rangle = |\uparrow\downarrow\downarrow\rangle.$$

Matrix elements do not change under the JWT, since it is only a rewriting of operators and a renaming of basis states.

1.8 Proofs about the JWT

Claim: $\{Z_j, c_j^{(+)}\} = 0$.

Proof. On the single site j , with $Z_j |0\rangle_j = |0\rangle_j$, $Z_j |1\rangle_j = -|1\rangle_j$, $c_j |0\rangle_j = 0$ and $c_j |1\rangle_j = |0\rangle_j$,

$$Z_j c_j |0\rangle_j = Z_j \cdot 0 = 0 = c_j |0\rangle_j = c_j Z_j |0\rangle_j, \quad (12)$$

$$Z_j c_j |1\rangle_j = Z_j |0\rangle_j = |0\rangle_j = c_j |0\rangle_j, \quad (13)$$

$$c_j Z_j |1\rangle_j = -c_j |1\rangle_j = -|0\rangle_j. \quad (14)$$

Hence, on both basis states $\{Z_j, c_j\} = Z_j c_j + c_j Z_j = c_j - c_j = 0$, and since $Z_j^\dagger = Z_j$ it follows that $\{Z_j, c_j^\dagger\} = 0$ as well. (see Coleman, Chapter 4.1) $\square$

Claim: $[a_i^{(+)}, a_j^{(+)}] = 0 \ \forall i, j$.

Proof. Assume w.l.o.g. that $i < j$. Using $(*)$,

$$\begin{aligned} a_j a_i &= (Z_1 \cdots Z_{j-1} c_j) (Z_1 \cdots Z_{i-1} c_i) \\ &\stackrel{\substack{[Z_j, c_\ell]=0, j \neq \ell \\ \{c_i, c_j\}=0}}{=} Z_1^2 \cdots Z_{i-1}^2 Z_i \cdots Z_{j-1} c_j c_i = -Z_i \cdots Z_{j-1} c_i c_j \quad (= 0 \text{ if } i = j), \end{aligned} \quad (15)$$

and likewise

$$\begin{aligned} a_i a_j &= (Z_1 \cdots Z_{i-1} c_i) (Z_1 \cdots Z_{j-1} c_j) \\ &\stackrel{[Z_i, c_\ell]=0, i \neq \ell}{=} Z_{i+1} \cdots Z_{j-1} c_i Z_i c_j \stackrel{\{Z_i, c_i\}=0}{=} -Z_i \cdots Z_{j-1} c_i c_j. \end{aligned} \quad (16)$$

Comparing the two, $a_i a_j = a_j a_i$, i.e. $[a_i, a_j] = 0$. Analogously for $[a_i^\dagger, a_j^\dagger]$ and for $i > j$; the case $i = j$ is straightforward since $c_i^2 = 0$. (i) $\square$

Claim: $[a_i, a_j^\dagger] = 0 \ \forall i \neq j$.

Proof. We can show it w.l.o.g. for $i < j$. Using $(*)$ and $(\star_2)$,

$$\begin{aligned} a_i^\dagger a_j &= (c_i^\dagger Z_{i-1} \cdots Z_1) (Z_1 \cdots Z_{j-1} c_j) \stackrel{Z_i^2=1, i < j}{=} c_i^\dagger Z_i \cdots Z_{j-1} c_j \\ &= Z_{i+1} \cdots Z_{j-1} c_i^\dagger Z_i c_j \stackrel{\{Z_i, c_i^\dagger\}=0}{=} -Z_i \cdots Z_{j-1} c_i^\dagger c_j = +Z_i \cdots Z_{j-1} c_j c_i^\dagger, \end{aligned} \quad (17)$$

and

$$a_j a_i^\dagger = (Z_1 \cdots Z_{j-1} c_j) (c_i^\dagger Z_{i-1} \cdots Z_1) \stackrel{[Z_j, c_\ell^{(+)}]=0, j \neq \ell}{=} Z_i \cdots Z_{j-1} c_j c_i^\dagger \underbrace{Z_{i-1}^2 \cdots Z_1^2}_{=1} = Z_i \cdots Z_{j-1} c_j c_i^\dagger. \quad (18)$$

Both reduce to the same expression, so $a_j a_i^\dagger - a_i^\dagger a_j = 0$ for $i < j$, i.e. $[a_j, a_i^\dagger] = 0$; the case $i > j$ follows straightforwardly. (ii) $\square$

Claim: $[a_i, a_i^\dagger] = 1 - 2n_i$ **with** $n_i = c_i^\dagger c_i$.

Proof. Using $(*)'$ and $Z_k^2 = \mathbb{1}$,

$$a_i^\dagger a_i = c_i^\dagger Z_{i-1} \cdots Z_1 Z_1 \cdots Z_{i-1} c_i = c_i^\dagger c_i = n_i, \quad (19)$$

$$a_i a_i^\dagger = Z_1 \cdots Z_{i-1} c_i c_i^\dagger Z_{i-1} \cdots Z_1 = c_i c_i^\dagger = 1 - c_i^\dagger c_i, \quad (20)$$

so that

$$[\hat{a}_i, \hat{a}_i^\dagger] = a_i a_i^\dagger - a_i^\dagger a_i = (1 - c_i^\dagger c_i) - c_i^\dagger c_i = 1 - 2\hat{n}_i. \quad (\text{iii})$$

□